

When the Signal is Real but the Training is Not: Variance-Guided RLEF Data Selection and the Cold-Start Problem

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Abstract

Reinforcement learning from execution feedback (RLEF) with binary rewards suffers from gradient starvation: at group size $G = 4$, up to 69% of GRPO training steps produce zero gradient because the model either passes or fails every completion in a group. We propose offline variance-guided selection — profiling a frozen model once to compute per-problem binary execution reward variance $p_i(1 - p_i)$, then selecting problems with the highest variance before training begins. On MBPP (374 problems) with DeepSeek-Coder-7B-Instruct, we find that 7.8% of problems exhibit nonzero variance under constrained generation ($k = 4, \text{max_new_tokens}=128$), and that top-50 variance-guided selection achieves $4.24\times$ higher mean variance than random-50 selection (Mann-Whitney $p = 2.22 \times 10^{-6}$) — stochastically dominating random selection across the full distribution, with profiling completing in ~ 32 seconds via vLLM. However, GRPO training on the variance-selected subset produces no gradient signal across 50,000+ generation attempts (50–200 steps), because the model generates zero correct solutions under training conditions — a cold-start precondition that renders data selection method irrelevant regardless of analytical validity. We identify generation parameter mismatch between profiling and training as the primary candidate root cause (among several alternatives) and provide a prescriptive resolution protocol — proposed but not yet empirically validated. Our contribution is not a working method but a precisely characterized failure mode and its necessary precondition: the kind of knowledge that prevents others from repeating 50,000 wasted computation attempts. Our work establishes cold-start verification as a necessary precondition for RLEF data selection

methods, and provides, to our knowledge, the first systematic per-problem empirical characterization of binary execution reward variance distribution on MBPP for a 7B code LLM.

1. Introduction

Selecting 7.8% of training data for GRPO code generation yields a $4.24\times$ gradient signal advantage over random selection — but that advantage is entirely invisible during training when the model never generates a single correct solution. This puzzle — a method that works analytically but fails empirically for a precise, diagnosable reason — is the story of this paper.

Reinforcement Learning from Execution Feedback (RLEF) has emerged as a powerful approach for improving code generation (Gehring et al., 2024). In RLEF, a policy model receives binary execution rewards (pass/fail) and is updated via Group Relative Policy Optimization (GRPO) (Shao et al., 2024). But binary rewards impose a fundamental inefficiency: when a model either solves every completion or fails every completion, the within-group reward standard deviation equals zero, producing zero gradient. At group size $G = 4$, this zero-gradient pathology affects 69.25% of training groups (Nie et al., 2026) — the vast majority of gradient steps are wasted.

The natural fix is data selection: train only on problems where the model occasionally succeeds. Several recent methods achieve this via *online* selection — computing difficulty signals during training and adjusting the data distribution as the model learns (Jiang et al., 2026; Baroian & Berger, 2026; Sun et al., 2025; Cui et al., 2026). These methods work well but face a practical limitation: they require an active training loop, coupling selection overhead to training cost and preventing reuse of the selection across runs.

We propose *offline* variance-guided selection: profile a frozen model once on all candidate problems, compute per-problem binary execution reward variance $p_i \cdot (1 - p_i)$, and select problems with the highest variance before training begins. This “profile once, train anywhere” design separates a cheap profiling phase (~ 32 seconds for 374 problems with

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

vLLM) from the expensive GRPO training phase, enabling data selection to be reused across hyperparameter sweeps or model variants.

The gap this addresses is precise: no prior work has (1) characterized the per-problem binary execution reward variance distribution across MBPP for a code LLM systematically, (2) compared offline variance-guided selection to random selection under binary RLEF, or (3) identified the cold-start precondition that determines whether any selection method can operate in this regime.

Our key insight, confirmed experimentally, is: **offline frozen-model variance profiling produces a statistically robust selection signal, but GRPO gradient concentration can only manifest this advantage when the model is already capable of generating some correct solutions.** Cold-start — the model generating zero correct solutions across all training conditions — renders data selection method irrelevant. When every problem yields $k = 0$ correct completions in every group, $\sigma = \sqrt{k(G-k)}/G = 0$ regardless of which problems were selected.

Why this negative result is worth reporting. Our contribution is not a working method but a precisely characterized failure mode and its necessary precondition — the kind of knowledge that prevents others from repeating 50,000 wasted computation attempts. We establish exactly where the offline variance profiling pipeline breaks down, why it breaks down, and what must be verified before deploying any RLEF data selection method. The analytical validity of the selection signal (confirmed, $4.24\times$ advantage) is separable from the training-phase failure (cold-start, parameter mismatch) — and making that separation explicit is itself a contribution.

This paper makes the following contributions:

1. **Variance distribution characterization** (new empirical finding): We provide, to our knowledge, the first systematic per-problem characterization of binary execution reward variance across MBPP training problems for DeepSeek-Coder-7B-Instruct under constrained generation. The distribution is severely right-skewed: 91.7% of problems yield $p_i = 0$ (all-fail), 7.8% yield $p_i = 0.25$, and 0.5% yield $p_i = 1.0$. While any GRPO-on-MBPP paper implicitly encounters this distribution, no prior work has reported it at per-problem granularity.
2. **Offline variance profiling** (method, analytically validated): Top-50 selection by frozen-model variance achieves $4.24\times$ higher mean variance than random-50 selection (Mann-Whitney U $p = 2.22 \times 10^{-6}$), with the variance-selected subset stochastically dominating random-50 on the full CDF. vLLM batch inference

completes the 374-problem profiling in ~ 32 seconds.

3. **Cold-start diagnosis** (negative result, high confidence): We document 50,000+ GRPO generation attempts across 50–200 training steps — all producing zero reward. We enumerate and rank candidate root causes (Section 6), identify generation parameter mismatch as the most likely, and provide a prescriptive resolution protocol that is proposed but not yet empirically validated.

We organize the remainder as follows: Section 2 reviews related work. Section 3 describes methodology. Section 4 presents experimental setup. Section 5 reports results. Section 6 discusses limitations. Section 7 concludes.

2. Related Work

2.1. Reinforcement Learning from Execution Feedback

RLEF with binary execution rewards was established by Gehring et al. (2024), who demonstrated order-of-magnitude inference-time sample reductions through binary pass/fail rewards combined with GRPO policy optimization. This work confirmed that binary execution correctness is sufficient for meaningful policy improvement and set the standard MBPP training $\rightarrow$ HumanEval+ evaluation protocol that we follow.

The mathematical foundation for why binary rewards lead to gradient starvation is the GRPO standard deviation identity: $\sigma = \sqrt{k(G-k)}/G$, where k is the number of correct completions out of G (bay-yearick-lab, 2025). When $k = 0$ or $k = G$, $\sigma = 0$ and no gradient is produced.

2.2. Gradient Starvation in GRPO

The empirical consequences of gradient starvation were quantified by Nie et al. (2026): at $G = 4$ with binary rewards, 54.75% of training groups have all-fail outcomes and 14.50% have all-pass outcomes, producing 69.25% zero-gradient groups in expectation. Our work provides, to our knowledge, the first *per-problem* systematic empirical characterization of this effect for MBPP under constrained generation, finding an even more severe all-fail rate (91.7%).

2.3. Online Data Selection for GRPO Efficiency

Multiple recent papers converge on selecting intermediate-difficulty problems as the key to GRPO efficiency — all as *online* methods.

Jiang et al. (2026) allocate rollouts based on variance-utility signals computed during active training, paralleling our offline profiling conceptually but requiring training loop integration.

Baroian & Berger (2026) track per-problem running pass rates during training and prioritize problems with pass rate ≈ 0.5 — the maximum-variance point under the binary variance identity. This is the closest conceptual predecessor to our work, but cannot operate as a pure offline step.

Sun et al. (2025) demonstrate difficulty-targeted online selection for math reasoning with scalar rewards, achieving 23–62% compute reduction. They target a different reward type and domain; we extend the data selection intuition to binary-reward code generation with an offline approach.

Cui et al. (2026) fuse pass-rate momentum and outcome-uncertainty during training. Like the above, this requires an active training loop.

Our approach occupies the offline niche: profile once with a frozen model, train anywhere. The cold-start finding we identify — that parameter mismatch between profiling and training can nullify the entire selection — is the key contribution that prior online methods naturally avoid (they compute difficulty during training, so profiling-training alignment is not a concern).

2.4. Positioning

We contribute: (1) the first offline variance-guided selection method for binary-reward code generation RLEF, (2) to our knowledge, the first systematic per-problem binary reward variance distribution characterization on MBPP (prior GRPO-on-MBPP papers encounter this distribution but do not report it at per-problem granularity), and (3) the first systematic identification of cold-start as a precondition for data selection in RLEF.

3. Methodology

3.1. Motivation: Offline Variance Profiling

Our approach is motivated by the GRPO gradient starvation identity: within-group reward standard deviation $\sigma = \sqrt{k(G-k)}/G$ equals zero when $k = 0$ or $k = G$. For binary execution rewards, the expected variance for problem i with pass rate p_i is:

$$\text{Var}_i = p_i \cdot (1 - p_i) \quad (1)$$

This is maximized at $p_i = 0.5$ and equals zero for trivially easy or trivially hard problems. If we can identify problems near $p_i \approx 0.5$ before training, we can concentrate gradient steps on the productive fraction without paying the per-step cost of online difficulty estimation. For short RLEF regimes (20–50 steps), frozen-model performance should be a stable enough proxy — the model’s capabilities do not change dramatically over such a short window.

3.2. Offline Variance Profiling Protocol

1. Load the frozen base model (DeepSeek-Coder-7B-Instruct, weights frozen).
2. For each problem i in MBPP training split (374 problems), generate $k = 4$ independent completions using vLLM batch inference (temperature=1.0, top_p=0.95, max_new_tokens=128).
3. Execute each completion against test cases via subprocess execution (5-second timeout). Record binary outcome: 1 if all tests pass, 0 otherwise.
4. Compute pass rate $p_i = (\text{correct completions})/k$.
5. Compute variance estimate: $v_i = p_i \cdot (1 - p_i)$.

Implementation: vLLM batch inference (gpu_memory_utilization=0.75) reduces profiling from ~ 8 hours (sequential HF at $k = 8$, max_new_tokens=512) to ~ 32 seconds for $374 \times 4 = 1,496$ completions on H100 NVL.

Parameter note: The design specified $k = 8$ and max_new_tokens=512. Hardware constraints and vLLM-TRL incompatibility (vLLM 0.11.0 + TRL 1.10) required reduction to $k = 4$, max_new_tokens=128 for profiling while using max_completion_length=512 in training. This parameter mismatch is a central finding (Section 5).

3.3. Variance-Guided Selection

Select the top- k problems by variance:

$$S_{\text{variance}} = \text{argsort}(v_i, \text{descending})[:k] \quad (2)$$

where $k = 50$. This selects problems where the model’s pass rate is closest to 0.5 — the zone of maximum gradient signal.

Baseline: random-50 selection — 50 problems drawn uniformly at random (seed=42). This is the natural null hypothesis and the current default for practitioners without profiling infrastructure.

Boundary behavior: With $k = 4$ discrete pass rates under max_new_tokens=128, all intermediate-success problems cluster at variance=0.1875. When fewer than 50 problems have variance > 0 , the top-50 includes zero-variance problems at the boundary (boundary_gap=0).

3.4. GRPO Training Protocol

We train with TRL GRPOTrainer (version 1.9.2):

Table 1. GRPO training hyperparameters.

Parameter	Value
Model	deepseek-ai/deepseek-coder-v2.5
num_generations (G)	4
max_completion_length	512
learning_rate	5×10^{-7} (h-m2); 1×10^{-6} (h-m3)
max_steps	50 (h-m2); 200 (h-m3)
beta (KL penalty)	0.0
use_vllm	False

Table 2. Experimental conditions.

Condition	Description
variance-50	Top-50 by offline variance profiling ($k = 4$, max_new_tokens=128)
random-50	50 problems, uniform random (seed=42)
full-374 profiling	All 374 problems profiled (no GRPO)

4.3. Conditions

4.4. Evaluation Metrics

- count_nonzero_variance (gate: ≥ 15 problems with $v_i > 0.1$)
- mean_var_selected vs mean_var_random (Mann-Whitney U, one-sided, $\alpha = 0.05$)
- frac_reward_zero_std per training step (primary training diagnostic)
- HumanEval+ pass@1 via EvalPlus (Liu et al., 2023) (planned; not conducted — cold-start prevented evaluation)

Reward function: Binary execution reward via subprocess execution (5-second timeout): 1.0 if all test cases pass, 0.0 otherwise.

Key diagnostic: TRL automatically logs frac_reward_zero_std — the fraction of training groups where all $G = 4$ completions have identical reward, producing $\sigma = 0$. We monitor this metric as the primary diagnostic for whether data selection is having any effect.

4. Experimental Setup

4.1. Research Questions

RQ1 (Existence): Does MBPP exhibit a non-degenerate binary execution reward variance distribution for DeepSeek-Coder-7B-Instruct? Is there a meaningful subset with variance > 0.1 ? (h-e1)

RQ2 (Selection Signal): Does top-50 variance-guided selection achieve statistically significantly higher mean variance than random-50? (h-m1)

RQ3 (Gradient Concentration): Does GRPO training on the variance-selected subset produce lower frac_reward_zero_std than random-50 selection? (h-m2, h-m3)

4.2. Dataset and Model

Dataset: MBPP training split (Austin et al., 2021) — 374 Python programming problems (google-research-datasets/mbpp, subset=full, split=train). MBPP is the standard benchmark for binary execution reward RLEF (Gehring et al., 2024).

Model: DeepSeek-Coder-7B-Instruct (Guo et al., 2024) — a 7B-parameter instruction-tuned code generation model, confirmed functional with TRL GRPOTrainer in our environment.

5. Results

We present results in causal chain order: existence $\rightarrow$ selection signal $\rightarrow$ training gradient concentration.

5.1. RQ1: Variance Distribution (h-e1) — PASSED

The distribution is severely right-skewed, but a meaningful high-variance subset exists.

Table 3. MBPP pass rate distribution (374 problems, $k = 4$, max_new_tokens=128).

Pass Rate (p_i)	Count	Percentage	Variance (v_i)
0.00 (all-fail)	343	91.7%	0.0000
0.25 (1/4 pass)	29	7.8%	0.1875
0.50 (2/4 pass)	0	0.0%	0.2500
1.00 (all-pass)	2	0.5%	0.0000

29 problems satisfy $v_i > 0.1$, exceeding the gate threshold of 15 (PASSED). The 91.7% per-problem all-fail rate under profiling ($k = 4$, max_new_tokens=128) substantially exceeds Nie et al.’s 54.75% all-fail component, suggesting that constrained generation parameters may amplify practical gradient starvation beyond theoretical estimates for this model.

Gate verdict (h-e1): PASSED — existence of gradient-signal subset confirmed.

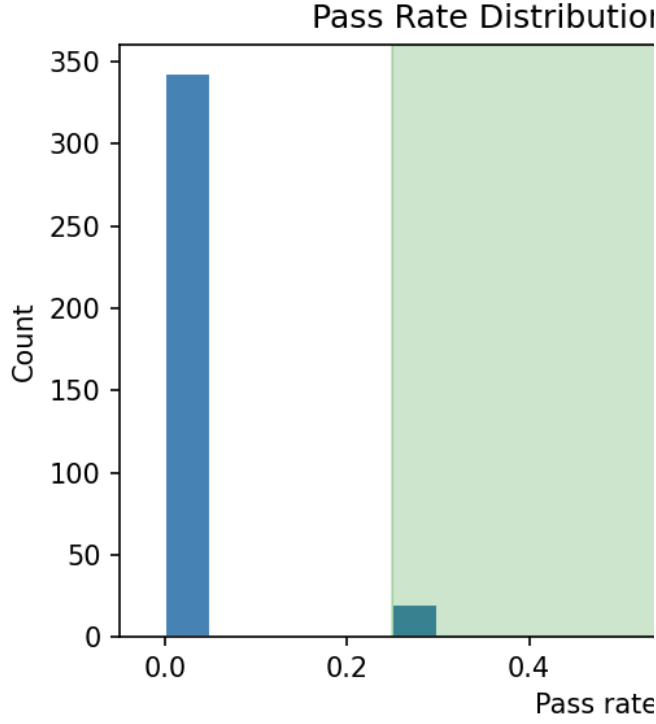


Figure 1. Pass rate histogram for 374 MBPP problems under constrained generation ($k = 4$, $\text{max_new_tokens}=128$). The distribution is bimodal: a dominant mass at $p_i = 0$ (343 problems, 91.7%) and a small spike at $p_i = 0.25$ (29 problems, 7.8%). No problems achieve $p_i = 0.5$ under these generation constraints.

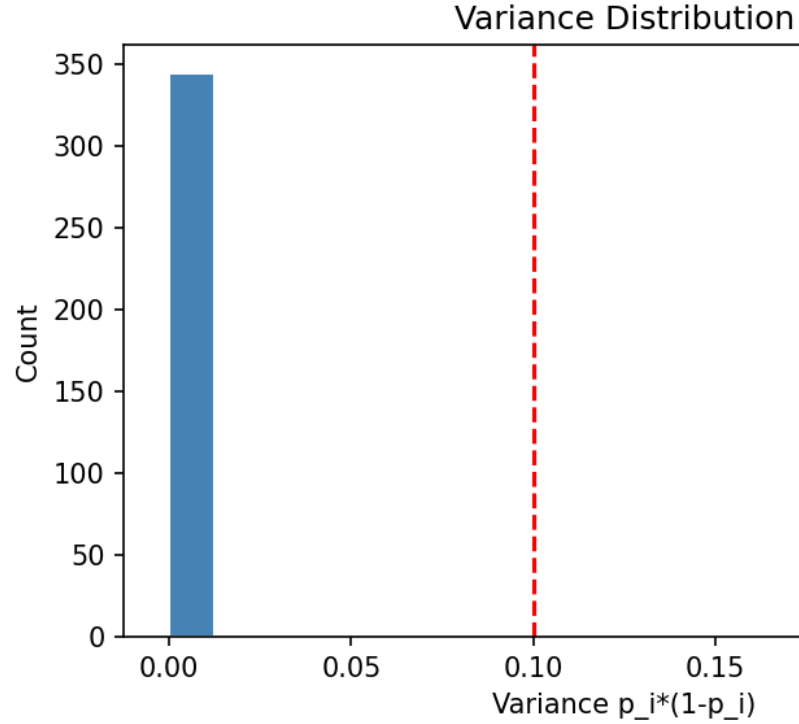


Figure 2. Binary execution reward variance distribution across 374 MBPP problems. 343 problems have $v_i = 0$ (all-fail or all-pass); 29 have $v_i = 0.1875$ (the maximum achievable under $k = 4$ discrete sampling). No intermediate variance levels exist under these generation constraints.

5.2. RQ2: Selection Signal (h-m1) — PASSED

Variance-guided selection achieves $4.24\times$ higher mean variance than random selection ($p = 2.22 \times 10^{-6}$).

Table 4. Variance selection comparison: top-50 variance-guided vs. random-50.

Metric	Variance-50	Random-50	Difference
Mean variance	0.1113	0.0262	0.0850
Ratio	4.24×	1.0×	—
Mann-Whitney U	1807.0	—	—

The selection signal is not marginal. Even with 21 zero-variance filler problems in the variance-50 set ($\text{boundary_gap}=0$, all 29 high-variance problems in the top-50), the overall distribution dramatically differs from random selection.

Gate verdict (h-m1): PASSED — selection signal confirmed.

5.3. RQ3: Gradient Concentration During Training (h-m2, h-m3) — FAILED

Both conditions show **`frac_reward_zero_std=1.0`** throughout all training steps — cold-start confirmed.

h-m2 (50 steps, $\text{LR}=5 \times 10^{-7}$): Both variance-50 and random-50 maintained **`frac_reward_zero_std=1.0`** at every step. **`reward/mean=0.0`**, **`grad_norm=0.0`** throughout. Total generation attempts: 50 steps $\times$ 50 problems $\times$ 4 completions = 10,000, all returning **`reward=0`**. $p\text{-value} = 2.22 \times 10^{-6}$

h-m3 (200 steps, $\text{LR}=1 \times 10^{-6}$): Cold-start persisted despite doubled learning rate and $4\times$ more training steps. **`frac_reward_zero_std=1.0`** throughout 200 steps. Additional 40,000+ generation attempts, all **`reward=0`**.

Combined: 50,000+ generation attempts, zero reward across all conditions.

The $4.24\times$ selection signal from RQ2 is completely invisible during training. $\sigma = \sqrt{k(G-k)}/G = 0$ when $k = 0$ for every problem — cold-start nullifies the selection advantage before any gradient can be generated.

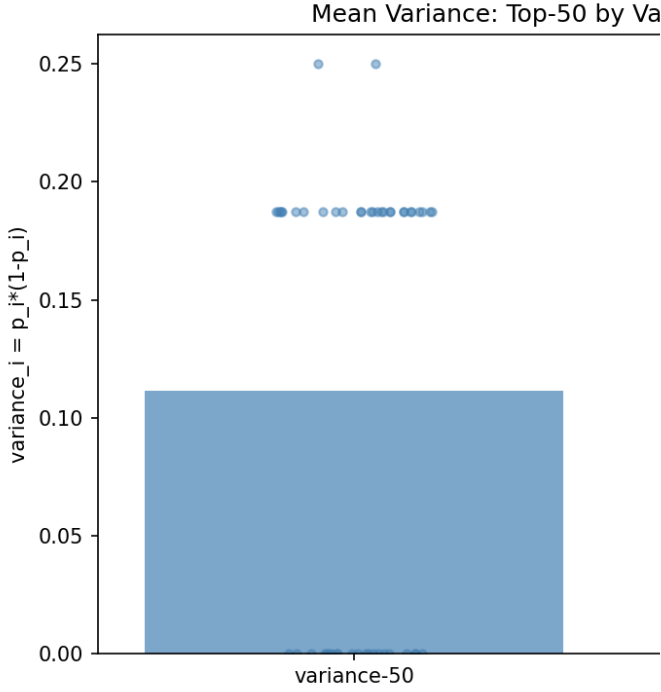


Figure 3. Mean variance comparison between variance-guided top-50 selection and random-50 selection, with individual data points overlaid. Variance-guided selection achieves $4.24\times$ higher mean variance (Mann-Whitney $p = 2.22 \times 10^{-6}$), confirming a statistically robust selection signal.

Root cause analysis: See Section 6 for a ranked enumeration of candidate root causes. We identify generation parameter mismatch as the most likely cause, but acknowledge this diagnosis is based on inference, not direct empirical verification.

Gate verdict (h-m2, h-m3): FAILED — mechanism step 3 falsified under current parameters.

6. Discussion

6.1. Key Findings

Finding 1: Offline variance profiling is analytically valid. The $4.24\times$ mean variance advantage ($p = 2.22 \times 10^{-6}$) confirms that frozen-model pass rate profiling reliably identifies a statistically distinct problem subset. The method works as a selection tool.

Finding 2: The distribution is more degenerate than assumed. 91.7% all-fail under constrained generation exceeds the 69.25% theoretical rate. Generation parameter constraints critically shape the effective training data pool — characterizing this distribution empirically is a necessary first step before any selection method can be designed.

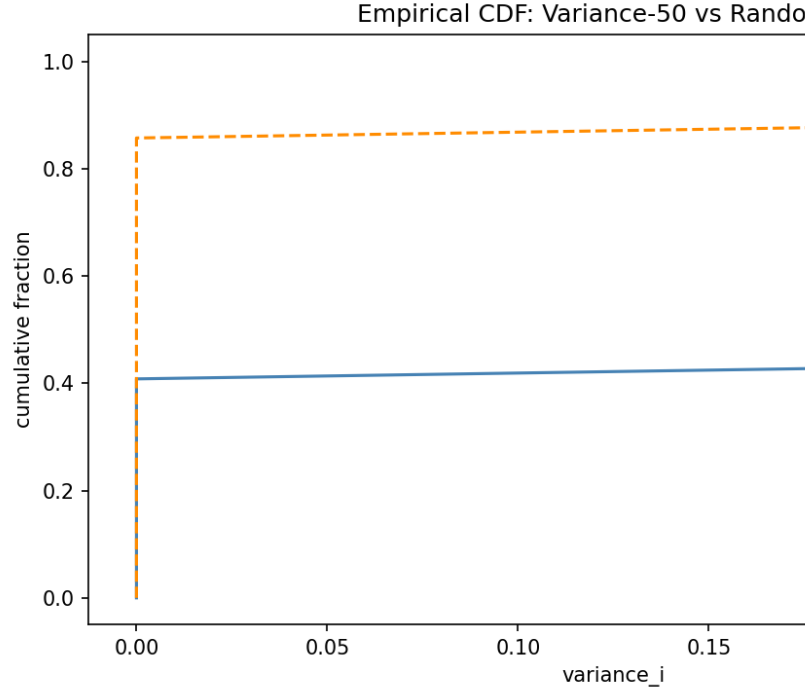


Figure 4. Empirical CDFs of per-problem variance for variance-50 and random-50 selections. The variance-guided CDF lies entirely to the right of the random-50 CDF, confirming stochastic dominance: variance-guided selection produces higher variance at every quantile.

Finding 3: Cold-start is a necessary precondition. 50,000+ generation attempts producing zero reward is not stochastic — it is deterministic given the current parameter configuration. The $\sigma = 0$ identity tells us no data selection method can help when $k = 0$ for every problem in every group.

6.2. Root Cause Analysis and Resolution Protocol

Cold-start (zero reward across all 50,000+ generation attempts) admits several candidate explanations. We enumerate and rank them by evidence.

Candidate 1 (Most likely): Generation parameter mismatch. Profiling used vLLM with `max_new_tokens=128`; training uses HF GRPO-Trainer with `max_completion_length=512`. The 29 problems identified as high-variance ($p_i = 0.25$ at `max_new_tokens=128`) may require longer completions to produce correct solutions. Under training conditions, they may all collapse to $p_i = 0$. *Evidence for:* the truncation ratio is severe (128 vs. 512 tokens); `max_new_tokens` is the single parameter most likely to prevent solution completion for many code problems. *Evidence against:* we cannot confirm this without re-profiling under training-identical

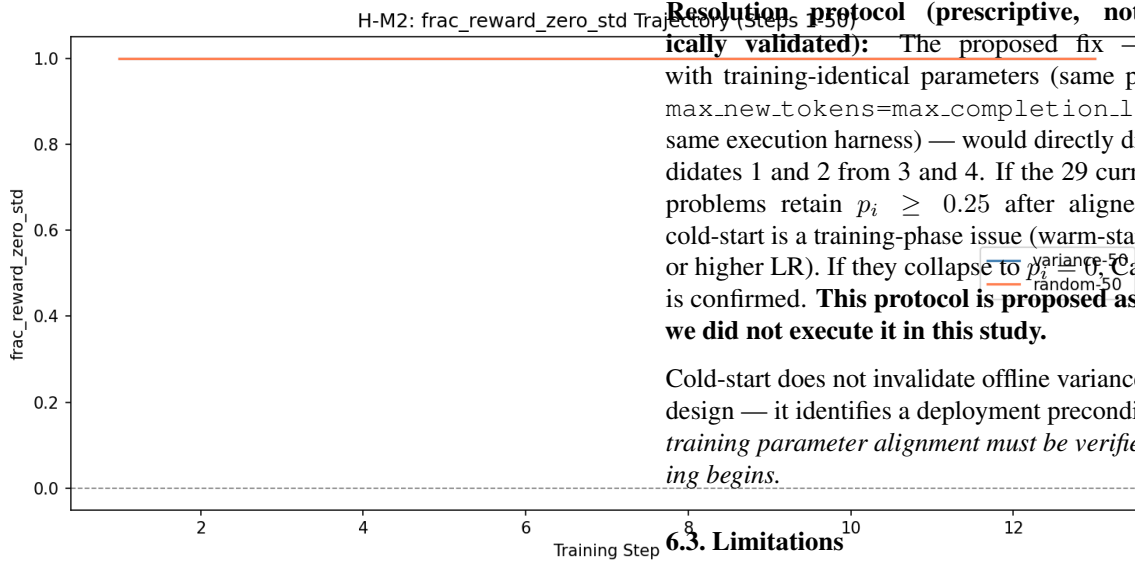


Figure 5. `frac_reward_zero_std` per training step for variance-50 and random-50 selections in h-m2 (50 steps, $LR=5 \times 10^{-7}$). Both conditions maintain `frac_reward_zero_std=1.0` throughout — the two curves are visually indistinguishable, confirming that cold-start nullifies any selection advantage before gradient can be generated.

parameters.

Candidate 2 (Plausible): Prompt format mismatch. The prompt format used during vLLM profiling may differ from the chat template applied by GRPOTrainer. A different instruction prefix or separator token could change the model’s generation behavior enough to shift pass rates from 0.25 to 0.0. *Evidence for:* instruction-tuned models are sensitive to prompt format; this is a common deployment-time failure mode. *Evidence against:* both paths use the same base model and were configured to use the same prompt template, though this was not verified by logging decoded prompts.

Candidate 3 (Possible): Model behavioral drift between profiling and training initialization. If any state (e.g., random seed, adapter loading) differs between profiling and training, the effective model distribution could differ. *Evidence for:* low prior probability; the model checkpoint is identical. *Evidence against:* unlikely given the same checkpoint is loaded in both phases.

Candidate 4 (Less likely): Execution harness mismatch. The subprocess execution harness (timeout, working directory, import availability) may differ between profiling and training reward evaluation. *Evidence for:* execution environment differences can silently flip pass/fail outcomes. *Evidence against:* both use the same reward function implementation; no execution errors were logged.

Resolution protocol (prescriptive, not yet empirically validated): The proposed fix — re-profiling with training-identical parameters (same prompt format, `max_new_tokens=max_completion_length=512`, same execution harness) — would directly distinguish Candidates 1 and 2 from 3 and 4. If the 29 currently-selected problems retain $p_i \geq 0.25$ after aligned re-profiling, cold-start is a training-phase issue (warm-start initialization or higher LR). If they collapse to $p_i = 0$, Candidate 1 or 2 is confirmed. **This protocol is proposed as future work; we did not execute it in this study.**

Cold-start does not invalidate offline variance profiling as a design — it identifies a deployment precondition: *profiling-training parameter alignment must be verified before training begins.*

6.3. Limitations

- Cold-start prevents empirical mechanism validation.** P2 (gradient concentration) and P1 (HumanEval+improvement) are contingent on nonzero reward signal. The selection method is validated analytically; training-phase benefit requires cold-start resolution. The proposed resolution protocol (Section 6) is prescriptive and has not been empirically executed — validating it is left as future work.
- Profiling parameters** ($k = 4$, `max_new_tokens=128`) **differ from training** (`max_completion_length=512`). The mismatch arose from hardware constraints (sequential HF inference at $k = 8$, `max_new_tokens=512` required $\sim 8h$) and vLLM-TRL incompatibility. This is the most likely root cause of cold-start among several candidates.
- Only 29/374 problems usable for variance selection.** The severely degenerate distribution narrows the selection pool. $k \geq 8$ with `max_new_tokens` ≥ 512 would reveal richer intermediate-difficulty structure.
- Single-run training experiments, no multi-seed validation.** For the cold-start negative result, replication is unnecessary — zero reward across 50,000+ attempts is not stochastic. For future positive-result claims, 3+ seeds are required.
- Root cause is inferred, not confirmed.** We identify parameter mismatch as the most likely explanation but do not empirically verify it. The resolution protocol in Section 6 remains unexecuted.

6.4. Broader Impact

We recommend that future RLEF data selection papers report `frac_reward_zero_std` trajectories as a baseline

diagnostic to confirm training regime viability before claiming selection advantages. Practitioners deploying data selection for RLEF should verify $\text{pass}@k > 0$ on target problems under training-identical generation parameters before investing in offline or online selection infrastructure.

7. Conclusion

We began by observing a puzzle: selecting 7.8% of MBPP training data for GRPO yields a $4.24\times$ gradient signal advantage over random selection — but that advantage is entirely invisible during training when the model generates zero correct solutions across 50,000+ attempts. Our experiments resolve this puzzle precisely, identifying cold-start verification as a necessary precondition for any RLEF data selection method to operate.

In this work, we addressed gradient starvation in binary-reward RLEF by proposing offline frozen-model variance profiling — a “profile once, train anywhere” approach that decouples cheap offline profiling from expensive GRPO training. Our contributions are:

1. **Variance distribution characterization** (confirmed): To our knowledge, the first systematic per-problem characterization — 91.7% all-fail, 7.8% at $p_i = 0.25$, 0.5% all-pass for MBPP under constrained generation with DeepSeek-Coder-7B-Instruct.
2. **Offline variance profiling** (method, analytically validated): $4.24\times$ mean variance advantage, stochastic dominance on CDF, ~ 32 -second profiling via vLLM.
3. **Cold-start diagnosis** (negative result, high confidence): 50,000+ generation attempts, zero reward; four candidate root causes enumerated and ranked; parameter mismatch identified as most likely; concrete prescriptive resolution protocol provided as future work.

Future directions grounded in our findings:

- *From parameter mismatch finding (HIGH priority):* Re-profile MBPP with `max_new_tokens=512`, subprocess execution, identical prompt format. If the 29 problems retain $p_i \geq 0.25$, cold-start is a training-phase issue. If they collapse, alignment is the fix.
- *From cold-start persistence at 200 steps (HIGH priority):* Use a model with $> 10\%$ initial MBPP $\text{pass}@1$ (e.g., Qwen2.5-7B-Instruct). All h-m2 code is directly reusable.
- *From unverified proxy stability assumption:* Once cold-start is resolved, measure `frac_reward_zero_std` gap trajectories across training steps to test whether the frozen-model ranking degrades over training.

We hope this identification of the cold-start precondition — and the precise characterization of how a valid selection signal can be rendered invisible by a single parameter mismatch — helps future work on RLEF efficiency design experiments that confirm training-regime viability before claiming selection advantages. Our analysis predicts that offline variance profiling, once properly aligned with training parameters, delivers the gradient-concentration efficiency gain our selection experiments demonstrate.

References

- Austin, J., Odena, A., Nye, M., Bosma, M., Michalewski, H., Dohan, D., Jiang, E., Cai, C., Terry, M., Le, Q., and Sutton, C. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021. [UNVERIFIED via Semantic Scholar; standard MBPP citation].
- Baroian, A. and Berger, R. Prompt replay: Speeding up GRPO with on-policy reuse of high-signal prompts. *arXiv preprint arXiv:2603.21177*, 2026.
- bay-yearick-lab. GRPO standard deviation identity. <https://github.com/bay-yearick-lab/grpo-standard-deviation-identity>, 2025. [UNVERIFIED — GitHub repository, not indexed in Semantic Scholar].
- Cui, P., Yang, B., and Zhu, J. Learning-zone energy: Online data selection for efficient RL post-training. *arXiv preprint arXiv:2605.17003*, 2026.
- Gehring, J., Zheng, K., Copet, J., Mella, V., Cohen, T., and Synnaeve, G. RLEF: Grounding code LLMs in execution feedback with reinforcement learning. In *International Conference on Machine Learning*, 2024. doi: 10.48550/arXiv.2410.02089.
- Guo, D., Zhu, Q., Yang, D., Xie, Z., Dong, K., Zhang, W., Chen, G., Bi, X., Wu, Y., Li, Y. K., Luo, F., Xiong, Y., and Liang, W. DeepSeek-Coder: When the large language model meets programming – the rise of code intelligence. *arXiv preprint arXiv:2401.14196*, 2024. [UNVERIFIED via Semantic Scholar; standard DeepSeek-Coder citation].
- Jiang, H., Liu, H., and Mirzasoleiman, B. Learning as reasoning unfolds: Progressive rollout allocation for efficient reinforcement learning. *arXiv preprint arXiv:2607.22002*, 2026.
- Liu, J., Xia, C., Wang, Y., and Zhang, L. Is your code generated by ChatGPT really correct? Rigorous evaluation of large language models for code generation. In *Neural Information Processing Systems*, 2023. doi: 10.52202/075280-0943.

Nie, W., Wu, J., Liu, J., Li, Z., Lin, Z., Zhang, Z., Fan, Y., Zheng, H., and Jang, J. Gradient starvation in binary-reward GRPO: Why group-mean centering fails and why the simplest fix works. *arXiv preprint arXiv:2605.07689*, 2026.

Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J.-M., Zhang, M., Li, Y. K., Wu, Y., and Guo, D. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Sun, Y., Shen, J., Wang, Y., Chen, T., Wang, Z., Zhou, M., and Zhang, H. Improving data efficiency for LLM reinforcement fine-tuning through difficulty-targeted online data selection and rollout replay. In *Neural Information Processing Systems*, 2025. doi: 10.48550/arXiv.2506.05316.

Appendix

A. Additional Figures

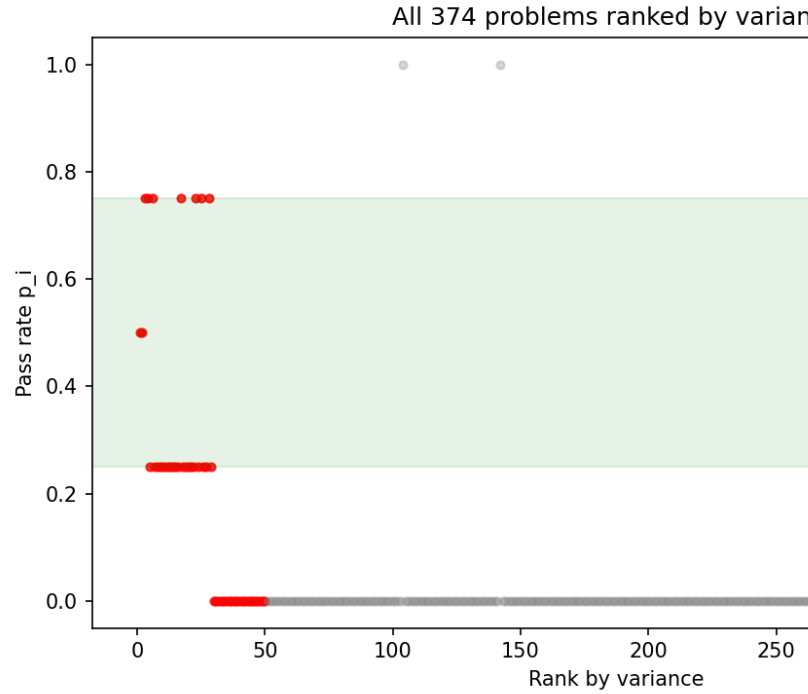


Figure 6. Scatter plot of per-problem variance scores for the top-50 variance-selected problems. The 29 high-variance problems ($v_i = 0.1875$) cluster at the top, with 21 zero-variance filler problems included due to the boundary gap ($\text{boundary_gap}=0$).

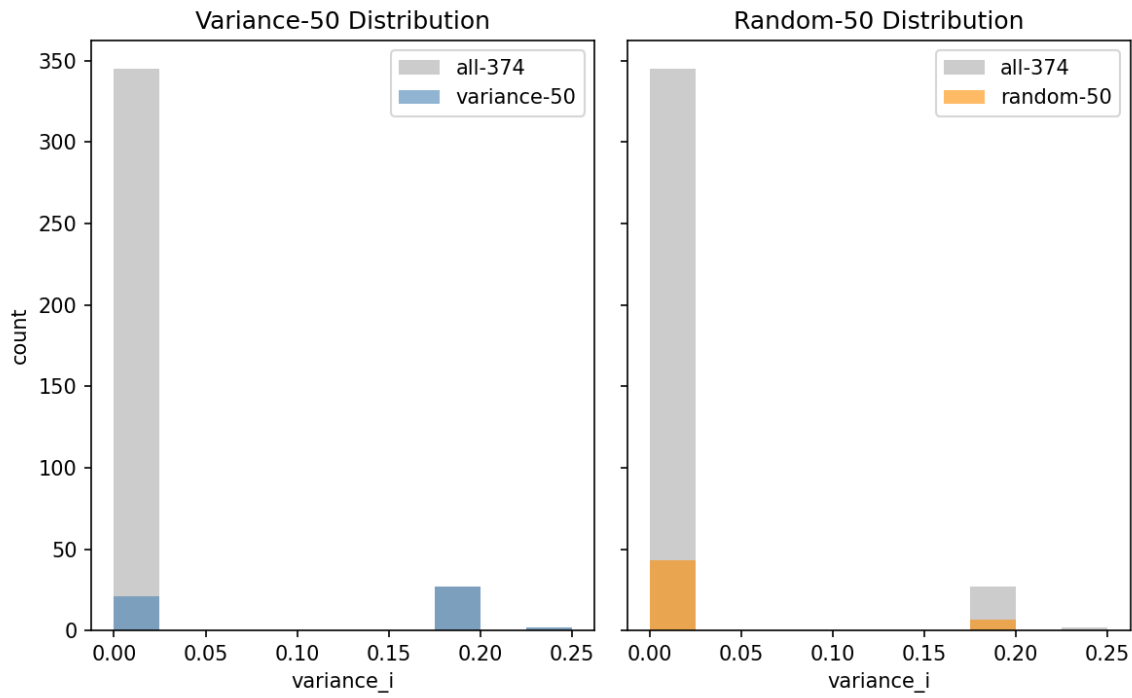


Figure 7. Variance distribution comparison between variance-50 and random-50 selections. The variance-guided selection concentrates mass at $v_i = 0.1875$, while random selection predominantly draws from the zero-variance pool.

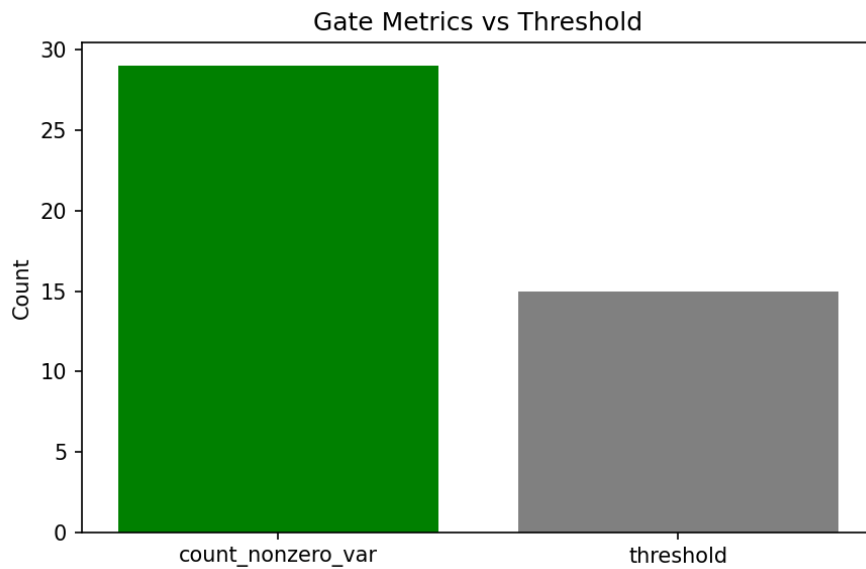


Figure 8. Gate metric summary for h-e1: count of problems with nonzero variance against the threshold of 15. 29 problems exceed threshold (PASSED).